2.6.2 Pharmacology Written Summary

PrP-lowering divalent siRNA 2439-exNA

Other names: 2439-exNA, 2439-s4, PrP-siRNA

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LIST OF ABBREVIATIONS

ASO	antisense oligonucleotide
CSF	cerebrospinal fluid
di-siRNA	divalent short interfering RNA
ED_{50}	median effective dose
exNA	extended nucleic acid
ICV	intracerebroventricular
IT	intrathecal
PrP	prion protein
siRNA	short interfering RNA

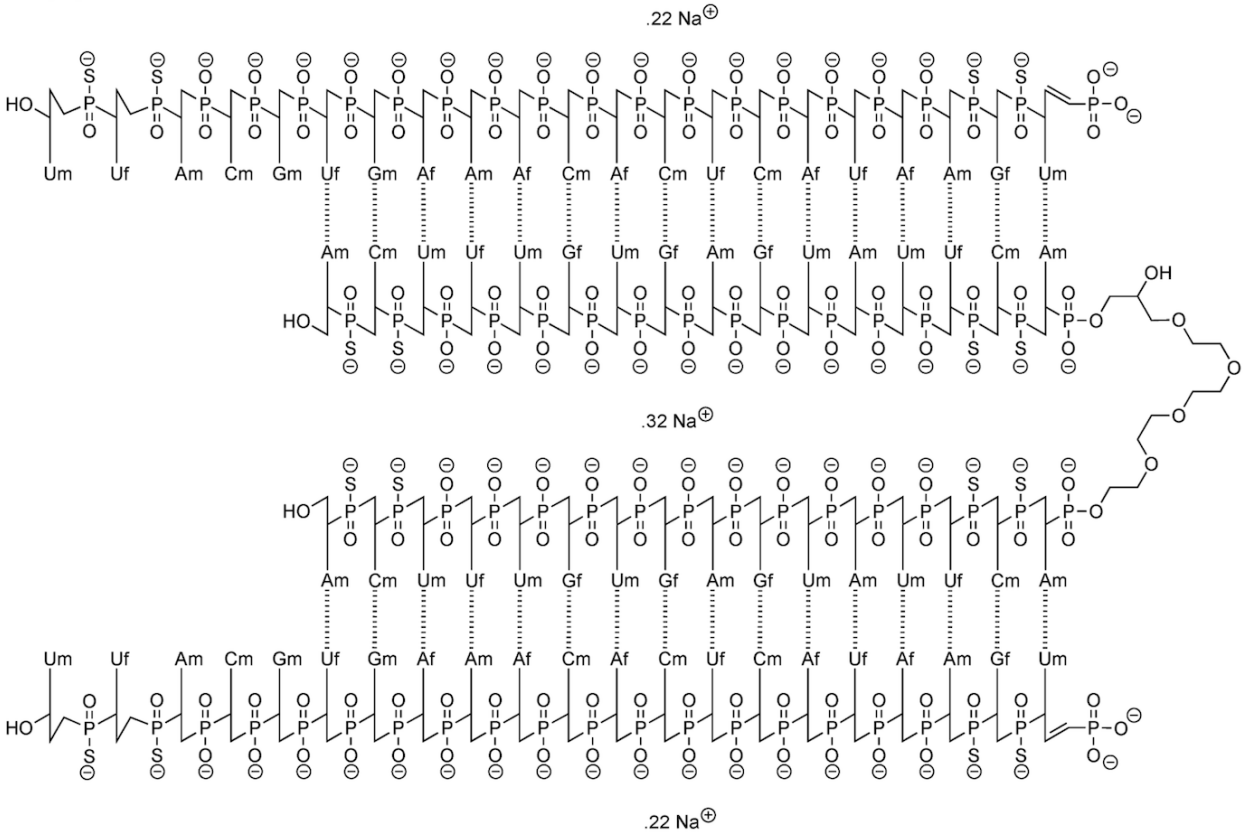
2.6.2 PHARMACOLOGY WRITTEN SUMMARY

2.6.2.1 Brief Summary

PrP lowering has already been well-established as a disease-modifying therapeutic strategy in prion disease through both genetic experiments and proof-of-concept experiments using antisense oligonucleotides (ASOs)[1,2]. In experiments described in detail in 4.2.1.1. Primary Pharmacodynamics and summarized briefly below, we further validated this therapeutic hypothesis by demonstrating that PrP lowering by a tool compound divalent siRNA targeting the mouse *Prnp* gene is effective at extending survival time in prion-infected mice.

PrP-lowering divalent siRNA 2439-exNA (Figure 2.6.2.1) consists of two covalently linked siRNA sense strands, each hybridized to an siRNA antisense strand. All strands are fully chemically modified to promote cellular uptake and exonuclease resistance, increasing mechanism of action. Like all oligonucleotide drugs, divalent siRNA are taken up into endosomes where they form a slow-release depot that is gradually released into cytosol, resulting in sustained pharmacologic activity over the course of months. 2439-exNA is designed to be complementary to the human *PRNP* RNA, encoding the prion protein (PrP), which is the cause of both initiation and progression in all cases of prion disease regardless of subtype or etiology. 2439-exNA is intended to be dosed into cerebrospinal fluid (CSF). This would be accomplished by an intrathecal (IT) route in human and accordingly IT was used as the delivery route in our GLP toxicology studies. In proof-of-concept studies in mice, given the difficulty of IT delivery in this smaller animal, we utilized intracerebroventricular (ICV) delivery.

Figure 2.6.2.1. Structure of 2439-exNA



2.6.2.2 Primary Pharmacodynamics

PrP-lowering divalent siRNA 2439-exNA was discovered through screening of siRNAs in human cultured cells and in transgenic mice expressing human PRNP to determine the siRNA sequences and chemical backbones providing the deepest lowering of the PRNP RNA transcript and PrP protein. A battery of non-GLP studies were performed at the Broad Institute in order to identify, prioritize, and characterize PrP-lowering siRNA 2439-exNA (also known as simply 2439-exNA or 2439-s4), as well as to test the efficacy of a surrogate divalent siRNA targeting the mouse *Prnp* gene in a disease model. All of these studies are described in detail in section 4.2.1.1 Primary Pharmacodynamics.

Prion disease is a fatal, incurable neurodegenerative disease caused by misfolding of prion protein (PrP), encoded in humans by the gene *PRNP*[1]. Lowering of PrP expression either by genetic manipulation or by treatment with antisense oligonucleotides (ASOs) is known to be efficacious against prion disease in animal models. 50% lowering of PrP increases both survival time and healthy life in prion-infected mice, but does not prevent symptom onset nor halt disease progression. Additional drug candidates should seek to reduce PrP expression to even lower levels.

Divalent siRNA is a novel oligonucleotide drug modality with promising potency, durability, and biodistribution data in preclinical models, inspiring us to seek in this technology a new drug candidate for

prion disease. Divalent siRNA consists of two covalently linked sense strands, each hybridized to an antisense strand which, when loaded into the RNA Induced Silencing Complex (RISC), catalyzes cleavage of the target RNA.

We identified a tool compound with a sequence specific to the mouse *Prnp* gene and used it to establish the efficacy of PrP-lowering divalent siRNA therapy in prion-infected mice. We utilized a standard disease model — wild-type mice inoculated intracerebrally with RML prions — and used intracerebroventricular (ICV) delivery of divalent siRNA into CSF. Treatment every 4 months, beginning at a disease stage with neuropathology but before frank symptoms, with a dose that lowers PrP to 49% residual, yielded a 2.7-fold increase in survival time (from 165 to 442 days on the median). Treatment with a single dose upon the onset of symptomatic disease indicated by weight loss yielded a 64% increase in survival time (from 164 to 270 days on the median).

We then identified a highly potent siRNA sequence, denoted 2439, against the human PrP gene. We determined that a chemical scaffold denoted s4, incorporating extended nucleic acid and a 3′ antisense tail unmatched to the RNA target, yields superior in vivo potency. We nominated a compound named 2439-s4, also known as PrP-lowering divalent siRNA 2439-exNA, as our drug candidate. In dose-response experiments in transgenic mice expressing human PrP, we find that the median effective dose (ED_{50}) — the dose resulting in 50% lowering of *PRNP* transcript by RT-qPCR — of this drug candidate ranges from 5 µg to 18 µg across the 6 mouse brain regions tested.

2.6.2.3 Secondary Pharmacodynamics

We performed an in silico assessment of the potential hybridization-mediated off-target effects of 2439-exNA, described in detail in 4.2.1.2 Secondary Pharmacology. The reverse complement of bases 2-17 of the 2439-exNA sequence — CACTTTGTGAGTATTC — was searched in NCBI Nucleotide BLAST (https://blast.ncbi.nlm.nih.gov/Blast.cgi) with the following settings. Database: Standard databases (nr etc.): > RefSeq Select RNA sequences (refseq_select). Organism: Homo sapiens (taxid:9606). Optimize for: Highly similar sequences (megablast). Max target sequences: 5,000. Match/mismatch scores: 1,-1. All other settings default. The top 100 hits from BLAST were downloaded as an XML file and parsed into tabular form using a Python script generated by ChatGPT. The table was then filtered using a custom R script to include only matches where the search string was antisense to the target, consistent with the siRNA mechanism, and was sorted by number of mismatches. No sequences with fewer than 2 mismatches, other than the intended target *PRNP*, were found as potential binding mRNA sites, indicating a minimal risk of off-target mRNA binding.

The 16 bases from 2-17 are most critical for determining activity against the target RNA. This critical region of 2439-exNA is GAAUACUCACAAAGUG. The reverse complement of this, CACTTTGTGAGTATTC, is the target sequence present in the coding (sense) strand of *PRNP* DNA corresponding to the 3'UTR of the mRNA. This sequence was searched against the Homo sapiens RefSeq Select mRNAs using NCBI Nucleotide BLAST (https://blast.ncbi.nlm.nih.gov/Blast.cgi). This search confirmed *PRNP* as the sole gene with 0 mismatches to 2439-exNA. No genes had exactly 1 mismatch. 2 genes had 2 mismatches each: *FURIN* and *ZKSCAN3*. All other genes had at least 3 mismatches.

Overall, evidence in vivo from the CNS indicates that transcripts with ≥2 mismatches are unlikely to experience any off target lowering[3,4]. In the human transcriptome, all non-*PRNP* genes have at least 2 mismatches to 2439-exNA, thus minimizing the risk of unwanted off-target effects.

2.6.2.4 Safety Pharmacology

No stand-alone, safety pharmacology studies have been performed with 2439-exNA. However, appropriate safety pharmacology endpoints eg, cardiovascular, respiratory functions, and blood pressure measurements were included in the single dose GLP dog toxicology study, and neurobehavioral function (functional observation battery) was included in the single dose GLP rat toxicology study (Module 4, studies A-703-02 and A-703-01, summarized in Module 2, Section 2.6.6). No test article related effects were observed for the safety pharmacology endpoints measured.

In dogs, there were no significant changes in mean arterial pressure, heart rate, respiratory rate, body temperature, or electrocardiographic activity observed following up to 200 mg single dose IT, at 2349-exNA, with plasma exposure of up to 26,654 ng/mL at the 0.5 hour timepoint.

In rats, there were no 2349-exNA related changes observed in clinical signs or neurobehavior (functional observation battery) following up to 3 mg single dose IT.

Overall, 2349-exNA is considered to have very low likelihood of presenting safety risk for the cardiovascular, central nervous system (CNS) and respiratory systems at the proposed dose levels for the phase I trial.

2.6.2.5 Pharmacodynamic Drug Interactions

No studies were performed.

2.6.2.6 Discussion and Conclusions

Prion disease is a fatal, incurable neurodegenerative disease caused by misfolding of prion protein (PrP), encoded in humans by the gene *PRNP*[1]. Divalent siRNA can potently and durably lower PrP and is effective against prion disease.

In a disease model, wild-type mice infected with RML prions, administration of a surrogate divalent siRNA targeting the mouse *Prnp* gene resulted in an extension of healthy life in prion-infected animals treated pre-symptomatically, and in already-symptomatic animals treatment extended life without reversing existing weight loss. This is consistent with an inability of PrP lowering to address pre-existing neuronal loss, and with a lag time of a few weeks between target engagement at the RNA level and maximal lowering of PrP at the protein level. Our data support treatment of prion disease patients at both symptomatic and pre-symptomatic timepoints, while suggesting that the greatest good can be achieved in a pre-symptomatic preventive paradigm[2]

In PrP-lowering divalent siRNA 2439-exNA showed a depth of target suppression — as low as 17% residual whole hemisphere PrP after a single 348 µg dose and 14% with repeat dosing — never previously reported for PrP, with at least some activity out to 6 months after a single dose. The ED_{50} for the candidate in mice, measured by regional qPCR, ranges from 5 - 18 µg depending upon brain region, suggesting that a single dose of 2439-s4 could lower PrP by 50% in many human brain regions, which is important because prion disease is a whole brain disease.

The reverse complement of bases 2-17 of the 2439-exNA sequence was manually checked using the BLASTN algorithm in the NCBI database against human mRNA. No sequences with fewer than 2 mismatches, other than the target sequence *PRNP,* were found as potential binding mRNA sites, indicating a minimal risk of toxicity due to off-target mRNA binding.

No effects of 2439-exNA on cardiovascular, respiratory, or central nervous systems were noted in safety pharmacology evaluations.

The data from these nonclinical studies strongly support the safety and medical plausibility of 2439-exNA as a potential therapy for patients with prion disease.

2.6.2.7 Literature References

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